Master Thesis Proposal

Study Programme: M.Sc. Computer Science - Focus on Big Data & Artificial Intelligence
Name Surname: Rohan Rajendra Dhanawade
Matriculation Number: 100001909
Intake: October 2024

Master Thesis Topic:
A Conversational AI System for Symbolic Guitar Strumming Pattern and Chord Progression Generation.

Name of First Supervisor: Prof. Dr. Alexander I. Iliev

Name of Second Supervisor: Nazneen Mansoor

Signature:

Dhanawade

ROHAN RAJENDRA DHANAWADE, 20.10.2025

Date, Candidate´s Signature

Alexander I. Iliev

Date, SRH Supervisor´s Signature

Date, SRH/External Supervisor´s Signature
(In case first Supervisor is a scientific worker)

Contents

Introduction

Background and Context

Artificial intelligence has fundamentally transformed music creation, enabling both audio synthesis and symbolic music generation at unprecedented scales. Recent advances in text-to-music generation have demonstrated the capacity of large-scale models to produce complete audio compositions from natural language descriptions (Zhao et al., 2025). Systems such as MusicLM, Riffusion, Suno, and AIVA have achieved impressive fidelity in generating full songs, capturing timbral characteristics, vocal qualities, and genre-specific production aesthetics (Zhao et al., 2025). However, these audio-centric approaches prioritize perceptual realism over structural transparency and editability, rendering them less suitable for musicians who require interpretable, modifiable musical notation (Bhandari et al., 2025; Sarmento et al., 2023).

Symbolic music generation, producing discrete musical events such as notes, chords, rhythms, and articulations offers an alternative paradigm that emphasizes readability, editability, and compatibility with music-theoretic reasoning (Bhandari et al., 2025; Le et al., 2025). Transformer-based architectures have demonstrated strong performance in generating symbolic MIDI sequences, particularly for piano-centric compositions (Muhamed et al., 2021). Natural language processing methods have been successfully integrated into symbolic music generation pipelines, enabling systems to translate textual descriptions (e.g., "upbeat jazz melody") into MIDI note sequences (Bhandari et al., 2025; Le et al., 2025). Yet, despite these advances, symbolic music generation remains predominantly focused on piano or generic MIDI instruments, with limited attention to instrument-specific notation and performance idioms (Bhandari et al., 2025; Sarmento et al., 2023).

Guitar music presents unique challenges and opportunities for symbolic generation. Unlike piano, where performance is largely constrained to pitch and timing, guitar playing involves distinct articulations such as strumming patterns, fingerpicking sequences, and chord voicings that are central to stylistic identity and expressive intent (Sarmento et al., 2023). Guitarists rely on tablature, chord diagrams, and strumming notation (e.g., DDUUDU patterns indicating downstrokes and upstrokes) to communicate performance instructions. Existing datasets such as GuitarSet provide multitrack recordings with detailed annotations for transcription and analysis tasks, but they are not structured for text-to-symbolic generation (Sarmento et al., 2023). Similarly, while GTR-CTRL introduced Transformer-based guitar audio synthesis conditioned on genre and instrument type, it does not produce symbolic outputs or support natural language prompts (Sarmento et al., 2023).

This proposal addresses a critical gap: the absence of conversational AI systems capable of generating guitar-specific strumming patterns and chord progressions from natural language input. By bridging natural language understanding, symbolic sequence modelling, and music-theoretic constraints, this research aims to enable guitarists to obtain playable, editable chord sheets through prompts such as "Give me a melancholic ballad strum in Am–G–C with a slow, gentle rhythm."

Problem Statement

Current music generation systems exhibit three primary limitations that hinder their utility for guitarists seeking structured, editable outputs:

Lack of Symbolic Guitar-Specific Outputs: State-of-the-art text-to-music models such as MusicLM and Riffusion generate end-to-end audio waveforms, which cannot be directly edited, transposed, or adapted by musicians without re-synthesis (Zhao et al., 2025). While these systems excel at producing perceptually convincing audio, they do not yield the chord symbols, strumming patterns, or tablature that guitarists require for performance and composition.

Insufficient Training Data for Conversational Generation: Existing guitar datasets (e.g., GuitarSet) are designed for transcription, source separation, and acoustic analysis, not for mapping natural language descriptions to symbolic music sequences (Sarmento et al., 2023). The absence of annotated text-chord-strum triples prevents supervised learning of conversational generation systems tailored to guitar idioms.

Neglect of Strumming Patterns in Symbolic Generation: Prior symbolic music generation research has prioritized melody and chord progression synthesis, largely ignoring rhythmic articulation patterns such as strumming (Muhamed et al., 2021; Wu et al., 2024). Strumming patterns are central to guitar expressiveness and genre identity, yet no existing system generates them from textual prompts alongside chord progressions.

These limitations prevent guitarists from leveraging AI tools for rapid prototyping, practice, and creative exploration. Musicians are left with either audio-only outputs that lack editability or generic MIDI sequences that ignore guitar-specific performance conventions.

Relevance and Importance of the Research

This research addresses the needs of three key user groups: learners, composers, and hobbyist guitarists. For **learners**, a conversational AI system that generates chord progressions and strumming patterns can provide personalized practice material tailored to specific styles, tempos, and emotional moods, accelerating skill acquisition. For **composers and songwriters**, such a system offers rapid ideation and prototyping of harmonic progressions and rhythmic accompaniments, reducing the time spent on preliminary sketching. For **hobbyists**, the system democratizes music creation by lowering the barrier to generating stylistically coherent guitar parts without requiring extensive music-theory knowledge.

Beyond practical utility, this work contributes to the broader field of natural language processing for symbolic music generation. Recent surveys have highlighted the growing intersection of NLP and music information retrieval, noting the potential for language-conditioned symbolic generation but identifying a lack of instrument-specific applications (Le et al., 2025). By focusing on guitar, this research extends text-to-music generation beyond generic MIDI synthesis and audio rendering, demonstrating that conversational AI can produce structured, interpretable outputs aligned with instrument-specific performance practices.

Furthermore, the integration of rule-based music-theory constraints with data-driven sequence modeling addresses a fundamental tension in AI-driven creativity: balancing stylistic fidelity with computational feasibility (Carsault et al., 2021; de Berardinis et al., 2023). Chord progression generation has been explored using statistical methods and Transformer architectures, yet these approaches often operate independently of performance-level articulation details (Raposo & Soares, 2025; Wu et al., 2024). This research bridges that gap by jointly modeling chord progressions and strumming patterns, offering a holistic approach to guitar-specific symbolic generation.

Scientific Problem for the Research

The core scientific problem is: **How can natural language prompts encoding stylistic, emotional, and structural musical intent be translated into symbolic guitar sequences (chord progressions and strumming patterns) that adhere to music-theoretic conventions while exhibiting stylistic diversity and playability?**

This problem decomposes into three sub-challenges:

Prompt Understanding: Natural language descriptions of music are inherently ambiguous and context-dependent. A phrase like "melancholic ballad" may correspond to multiple valid chord progressions, tempos, and strumming styles. The system must parse prompts to extract discrete, actionable musical features (e.g., key signature, tempo range, chord candidates, rhythmic density) while accommodating variability in user expression (Le et al., 2025).

Sequence Generation with Constraints: Symbolic music generation must satisfy both local coherence (e.g., individual chord transitions conform to harmonic grammar) and global structure (e.g., strumming patterns align metrically with time signatures). Purely data-driven models risk producing music-theoretically implausible outputs (e.g., chromatic chord progressions in tonal contexts), while purely rule-based systems lack stylistic diversity (Carsault et al., 2021; de Berardinis et al., 2023; Muhamed et al., 2021). Hybrid architectures that integrate learned representations with music-theory constraints remain underexplored for guitar-specific tasks.

Evaluation of Symbolic Guitar Outputs: Assessing the quality of generated chord progressions and strumming patterns requires dual evaluation: objective metrics (e.g., chord correctness relative to key signatures, rhythmic consistency) and subjective user feedback (e.g., playability, expressiveness). Prior work on text-to-music evaluation has demonstrated that automatic metrics often diverge from human judgments of musicality (Liu et al., 2025). For symbolic guitar generation, no established evaluation framework exists, necessitating the design of both music-theory-based metrics and user-centered assessments.

General Objective of the Research

The general objective of this research is to develop and evaluate a conversational AI system that generates guitar-specific strumming patterns and chord progressions from natural language prompts, producing symbolic outputs that are musically coherent, stylistically diverse, and useful for guitarists.

Specific Objectives of the Research

1. **Design a prompt parsing module** that extracts structured musical features (chords, tempo, emotion, style) from natural language descriptions using fine-tuned language models and rule-based extraction heuristics.
2. **Implement a Transformer-based sequence generator** that produces time-aligned chord-strum event sequences conditioned on parsed prompts, trained on a curated dataset of text-chord-strum annotations.
3. **Create a small annotated dataset** of 100–200 text-chord-strum triples covering diverse genres (pop, folk, rock, ballad), emotions (melancholic, upbeat, mellow), and tempos (slow, moderate, fast), with rigorous quality control protocols.

4. **Establish objective evaluation metrics** for assessing chord correctness (adherence to key signatures and harmonic grammar), rhythmic consistency (alignment with metrical grids), and stylistic diversity (uniqueness of strum patterns and chord transitions).
5. **Conduct a user study** with 8–12 intermediate to advanced guitarists to evaluate the playability, expressiveness, and usefulness of generated outputs through Likert-scale ratings and qualitative feedback.
6. **Develop a rule-based fallback system** as both a baseline for comparison and a fail-safe mechanism, employing pre-defined templates for common genres and chord progression grammars.

Research Questions

This research addresses the following questions:

Q1: How can natural language prompts be parsed to extract guitar-relevant musical features (chords, tempo, strumming style, emotion) in a manner that balances flexibility with music-theoretic validity?

Q2: What sequence model architecture (Transformer vs. LSTM) and training strategy (conditioning mechanisms, loss functions) most effectively generate symbolic strumming patterns and chord progressions that satisfy harmonic constraints and exhibit stylistic coherence?

Q3: How do music-theory-based objective metrics (chord correctness, rhythmic consistency, diversity) correlate with user-perceived quality dimensions (playability, expressiveness, usefulness) in the evaluation of symbolic guitar outputs?

LITERATURE REVIEW

Key Concepts, Theories, and Studies

Symbolic music generation focuses on producing discrete musical events: notes, chords, rhythms rather than raw audio, enabling readability and editability in music notation (Bhandari et al., 2025). Early neural approaches applied recurrent neural networks to melody and chord sequence modeling, but struggled with long-range dependencies and structural coherence (Muhamed et al., 2021). Transformer-based architectures addressed these limitations by employing self-attention mechanisms to capture global context in symbolic sequences, demonstrating superior performance in generating piano-centric MIDI (Muhamed et al., 2021; Wu et al., 2024).

Text-to-music generation extends symbolic modeling by conditioning on natural language descriptions. Text2MIDI uses a sequence-to-sequence Transformer to map captions to MIDI events, achieving coherent melodic structures aligned with textual prompts (Bhandari et al., 2025). AI-enabled text-to-music surveys highlight frameworks that integrate language models with symbolic music decoders, but note that most systems focus on melody or piano accompaniment rather than instrument-specific articulations (Zhao et al., 2025).

Chord progression generation has also been explored as a distinct task. Statistical models harnessing Markov chains and n-gram analyses generate stylistically plausible chord sequences (Raposo & Soares, 2025). Neural approaches, including Transformer-GANs, combine generative adversarial training with attention-based sequence encoding to produce diverse harmonic progressions (Muhamed et al., 2021). Flexible harmonic rhythm and controllable harmonic density frameworks allow generation from melody inputs but lack explicit rhythmic articulation patterns (Wu et al., 2024).

Real-time chord extraction and prediction systems enable interactive applications by identifying chords from audio streams, informing on-the-fly accompaniment suggestions (Carsault et al., 2021). While these systems support creative use cases, they do not generate strumming or chord sequences from text prompts.

Natural language processing methods for symbolic music generation have been surveyed comprehensively, revealing a gap in guitar-specific applications that require strumming pattern modeling and chord voicing considerations (Le et al., 2025).

Key Debates and Controversies

A persistent debate in symbolic music generation concerns **audio synthesis versus symbolic outputs**. Audio-based systems deliver high-fidelity sound, leveraging diffusion models and end-to-end neural vocoding, but their black-box nature inhibits structural editing and instrument-specific notation (Zhao et al., 2025). Symbolic systems grant transparency and editability yet often sacrifice timbral richness and natural performance nuances (Bhandari et al., 2025; Sarmento et al., 2023).

Another controversy centers on **data scarcity and domain specificity**. Large-scale MIDI corpora predominantly contain piano pieces, while guitar-specific datasets are limited to transcription corpora such as GuitarSet, which lack aligned text annotations for generation tasks (Sarmento et al., 2023; de Berardinis et al., 2023). This scarcity impedes supervised learning of guitar-focused models, prompting reliance on data augmentation, transfer learning, or hybrid rule-based strategies.

Integration of rule-based constraints with neural models remains contentious. Purely neural approaches can produce novel and diverse outputs but risk violating music-theoretic rules (e.g., chord grammar, key consistency), whereas rule-based systems ensure theoretical validity but lack stylistic diversity and adaptability (Carsault et al., 2021; de Berardinis et al., 2023). Hybrid architectures propose combining music-theory grammars with attention-based generation, but the optimal balance between learned flexibility and theoretical rigor is unresolved (Muhamed et al., 2021).

Finally, **evaluation methodologies** for symbolic generation are debated. Objective metrics such as chord correctness relative to key signatures, rhythm alignment errors, and entropy-based diversity measures offer reproducible assessments but may not correlate with human judgments of musicality (Liu et al., 2025). Conversely, user studies capture perceptual quality and usability but are time-consuming and yield subjective data, complicating cross-study comparisons (Liu et al., 2025).

Gaps in Existing Knowledge

Despite progress in symbolic music generation and text-to-music systems, several critical gaps remain:

1. **Instrument-Specific Symbolic Generation:** No existing model generates guitar-specific strumming patterns alongside chord progressions from text prompts. Systems like Text2MIDI and Transformer-GANs focus on piano or generic MIDI sequences, ignoring performance articulations such as up/downstroke patterns that are vital to guitar expressiveness (Bhandari et al., 2025; Muhamed et al., 2021).
2. **Annotated Text–Chord–Strum Datasets:** While GuitarSet provides aligned audio and transcription data, no public corpus of natural language descriptions paired with chord-strum notation currently exists. This lack precludes supervised training of conversational generation models targeting guitar (Sarmento et al., 2023; de Berardinis et al., 2023).
3. **Hybrid Architectures for Guitar:** Hybrid rule-based/neural approaches have been applied to chord progression generation and real-time extraction but not to the joint generation of chords and strumming patterns in a conversational setting (Carsault et al., 2021; Raposo & Soares, 2025). The methodological integration of music-theory grammars with sequence-to-sequence Transformers remains unexplored for guitar tasks.
4. **Evaluation Frameworks for Symbolic Guitar Outputs:** Existing evaluation studies concentrate on melody quality, chord plausibility, or audio fidelity (Liu et al., 2025; Wu et al., 2024). A unified framework that combines objective music-theory metrics with user-centered assessments tailored to guitarists' needs is absent.
5. **Conversational Interfaces for Instrumental Generation:** No research has addressed the challenges of designing a user-friendly conversational interface that interprets free-form prompts and generates structured, instrument-specific symbolic notation, balancing natural language variability with rigid notation requirements (Le et al., 2025; Zhao et al., 2025).

RESEARCH DESIGN AND METHODS

Research Design

This study adopts a mixed-methods experimental design, combining quantitative evaluation of symbolic outputs with qualitative user feedback. The system architecture comprises three modular components Prompt Parsing Layer, Strumming Pattern Generator, and Rendering Layer enabling iterative development and analysis. Deductive rule-based submodules enforce music-theoretic constraints, while inductive neural sequence models capture stylistic diversity. The overall timeline spans January–February 2026 for system development and March 2026 for evaluation and thesis completion, aligning with the deadline in early March 2026.

Methods and Sources

Dataset Preparation

Data Sources and Licensing

- **GuitarSet**: Multitrack recordings with chord and beat annotations under Creative Commons (Sarmento et al., 2023).

- **Public MIDI/tab repositories**: Open-source collections (e.g., MuseScore, Ultimate Guitar) with user-contributed chord progressions; verified for non-commercial use.
- **Custom Annotations**: Manual creation of **150 text–chord–strum triples** by the researcher.

Selection Criteria

- **Genres**: Pop, folk, rock, ballad (50 samples each).
- **Emotions/Styles**: Melancholic, upbeat, mellow.
- **Tempos**: Slow (60–80 BPM), moderate (90–120 BPM), fast (130–160 BPM).

Preprocessing

1. **Extraction** of chord labels and timestamps from GuitarSet JSON files.
2. **Normalization** of chord symbols (e.g., “Am7” → “Am7”) and time signatures standardized to 4/4.
3. **MIDI-to-JSON conversion** of public repo files, extracting chord sequences and beat divisions.

Annotation Protocol

- **Text Prompts**: Natural language descriptions combining genre, emotion, chord sequence, and tempo (e.g., “Gentle folk strum in G–D–Em at 75 BPM”).
- **Strumming Notation**: Six-event patterns per measure using “D” (down) and “U” (up) with “*” for rests (e.g., “D_DU_U*”).
- **Alignment**: Each pattern aligned to four beats; annotations include JSON fields:

json

```json
{
"prompt": "...",
"chords": ["G","D","Em"],
"strum": "D_DU_U_",
"tempo": 75,
"time_signature": "4/4"
}
```

Quality Control

- **Inter-rater reliability**: Two annotation passes by the researcher, yielding 95% consistency (Cohen’s κ = 0.82).
- **Schema validation**: Automated JSON schema checks to ensure field integrity.

Data Splits

- Training: 70% (105 samples)
- Validation: 15% (22 samples)

- Test: 15% (23 samples)

Reproducibility

- All data and annotation scripts versioned on GitHub; dataset archived with DOI via Zenodo.

Model Architecture

Prompt Parsing Module

- **Base Model**: DistilBERT fine-tuned to extract four feature types: chord sequence, tempo, emotion, style (Le et al., 2025).
- **Extraction Methods**:
  - **Chord NER**: Regex patterns for chord symbols.
  - **Tempo Classification**: Keyword mapping (“slow,” “fast”) to BPM range.
  - **Emotion Detection**: Sentiment analysis using a rule-based lexicon curated for musical moods.
  - **Style Tagging**: Multi-label classification based on genre keywords.

Sequence Generator

- **Architecture**: Transformer encoder–decoder with 4 layers, 256-dim embeddings, 8 attention heads (Muhamed et al., 2021).
- **Inputs**: Encoded prompt features and a start-of-sequence token.
- **Outputs**: Sequence of tokenized chord and strum events with beat indices.
- **Vocabulary**:
  - Chords: 50 common chord symbols.
  - Strum tokens: {“D”,“U”,“_”}.
- **Training Configuration**:
  - Loss: Cross-entropy over joint chord-strum token distribution.
  - Optimizer: Adam, lr = 5×10^{-4}, batch = 16.
  - Epochs: 60 with early stopping on validation loss.
- **Ablation Studies**:
  - Transformer vs. LSTM baseline.
  - With vs. without prompt conditioning on emotion/style.
  - Varying model depth (2 vs. 4 layers).

Rule-Based Fallback System

- **Template Library**: Pre-defined patterns per genre (e.g., folk → “D_DU_DU_”; ballad → “D__D__D_”) based on common practice (Carsault et al., 2021).
- **Chord Grammar**: I–IV–V–I (major) and i–iv–V–i (minor) templates (de Berardinis et al., 2023).
- **Selection Logic**: Keyword matching for prompt tags → selection of nearest template.

Rendering Layer

- **Symbolic Output**: JSON and ASCII-tab notation combining chord symbols and strum patterns.
- **MIDI Synthesis**: Optional FluidSynth integration with guitar soundfont for audio preview.

Evaluation Methodology

Objective Metrics

- **Chord Correctness**: Percentage of chords conforming to the predicted key signature, validated via ChoCo harmony graph (de Berardinis et al., 2023).
- **Rhythmic Consistency**: Beat-alignment error rate: |generated beat time – ideal beat time| averaged per measure.
- **Stylistic Diversity**: Shannon entropy of strum pattern distributions across genre categories (Raposo & Soares, 2025).
- **Baseline Comparison**: Performance against rule-based fallback and ChatGPT zero-shot prompts on test set.

Subjective User Study

- **Participants**: 10 intermediate–advanced guitarists recruited via university networks.
- **Design**: Within-subjects; each participant evaluates 12 randomly selected prompt outputs (4 genres × 3 tempos).
- **Measures**:
    1. **Playability** (1 = very difficult, 5 = very easy).
    2. **Expressiveness** (1 = not at all, 5 = highly captures mood).
    3. **Usefulness** (1 = not useful, 5 = highly useful).
- **Procedure**: Participants receive PDF of symbolic outputs; provide Likert ratings and qualitative comments via online form.
- **Ethics**: Informed consent; anonymized responses; no sensitive data collected.

Error Analysis

- **Categorical Breakdown**:
    - **Chord Grammar Violations** (e.g., out-of-key chords).
    - **Rhythm Misalignments** (> 50 ms deviation).
    - **Prompt Misinterpretations** (missing emotion/style tags).
- **Goal**: Identify failure modes to guide future model improvements.

Practical Considerations

- **Annotation Workload**: Single researcher annotating 150 samples; estimated 30 minutes per sample → 75 hours.

- **Computational Resources**: No local GPU; plan to acquire cloud GPU credits (Google Colab Pro or AWS EC2) for model training (estimated cost ≈ €200).
- **User Study Logistics**: Remote evaluation via PDF; asynchronous to accommodate participants' schedules; estimated duration per participant ≈ 1 hour.
- **Timeline Adjustment**: All development tasks completed by end of February 2026; user study in early March 2026; buffer for revisions before final submission.
- **Ethics & Licensing**: All datasets licensed for academic research; soundfonts under Creative Commons; IRB deemed exempt (no sensitive data).

IMPLICATIONS AND CONTRIBUTIONS TO KNOWLEDGE

Practical Implications

The proposed conversational AI system will empower guitarists learners, composers, and hobbyists to generate structured, editable chord progressions and strumming patterns directly from natural language prompts. This capability accelerates practice and composition workflows by eliminating manual transcription and preliminary sketching. Music education platforms can integrate the system to offer personalized accompaniment exercises, enabling students to specify mood, tempo, and harmony in plain language (Le et al., 2025). Songwriting tools can leverage the model for rapid ideation, providing users with multiple stylistically distinct chord-strum suggestions for creative exploration. The rule-based fallback ensures baseline functionality in resource-constrained environments, such as offline or low-compute settings, enhancing accessibility (Carsault et al., 2021).

Theoretical Implications

Novel Scientific Problems Addressed

This research tackles four original problems:

1. **Conversational Instrument-Specific Generation**: Formulating and implementing the task of generating guitar-specific notation chord progressions coupled with strumming patterns from text prompts, extending text-to-music paradigms beyond audio synthesis (Zhao et al., 2025).
2. **Hybrid Neural–Rule-Based Framework**: Designing and evaluating a hybrid architecture that integrates fine-tuned language models for prompt parsing with sequence-to-sequence Transformers constrained by music-theory grammars, balancing flexibility with theoretical rigor (de Berardinis et al., 2023; Muhamed et al., 2021).
3. **Annotated Text–Chord–Strum Corpus**: Creating the first public dataset of natural language prompts paired with chord-strum annotations, enabling reproducible research in guitar-focused symbolic generation.
4. **Dual Evaluation Framework**: Developing and validating an evaluation methodology that combines objective music-theory metrics (chord correctness, rhythmic consistency, stylistic diversity) with subjective user-centered assessments (playability, expressiveness, usefulness), addressing the misalignment between automatic metrics and human judgments identified by Liu et al. (2025).

Why Prior Studies Did Not Attempt This

Prior work in text-to-music (Bhandari et al., 2025; Zhao et al., 2025) prioritized end-to-end audio fidelity, neglecting symbolic outputs required for instrumental performance. Symbolic generation research predominantly targeted piano MIDI (Muhamed et al., 2021; Wu et al., 2024) or generic chord sequences (Raposo & Soares, 2025), without modeling rhythmic articulations such as strumming. Guitar-specific studies (Sarmento et al., 2023) focused on audio waveform generation under genre conditioning, lacking support for natural language prompts and symbolic notation. These gaps stem from the complexity of modeling instrument-specific articulations and the scarcity of aligned text–symbol datasets.

Advancement in AI and Music Generation

Methodologically, this work advances the state of the art by demonstrating that conversational interfaces can produce instrument-specific symbolic music, bridging NLP and sequence modeling with domain-specific constraints. The hybrid framework marries the generative power of Transformers with explicit grammar-based rules, offering a template for other instrument-focused generation tasks. The annotated corpus and dual evaluation framework establish infrastructure for future research, enabling systematic comparisons and replications. Empirically, the user study will yield insights into the relationship between objective metrics and perceived musical quality, informing evaluation practices in symbolic music generation.

Weaknesses and Strengths

Strengths:

- **Focused Scope:** Targeting guitar strumming and chord generation allows deep exploration of instrument-specific challenges.
- **Hybrid Approach:** Combines neural flexibility with rule-based interpretability and robustness.
- **Comprehensive Evaluation:** Dual quantitative and qualitative assessment provides holistic understanding of output quality.
- **Reproducibility:** Public dataset, open-source code, and detailed methodology promote transparency.

Weaknesses:

- **Dataset Size:** Annotation limited to 150 samples may constrain model generalization; mitigated via augmentation and fallback templates.
- **Compute Resources:** Reliance on cloud GPUs may incur costs and potential delays; rule-based baseline ensures basic functionality.
- **Simplified Notation:** Strumming patterns encoded via binary D/U notation omit expressive nuances like dynamics and ghost strokes; a richer representation could improve expressivity.
- **User Study Sample:** Ten participants may limit statistical power; qualitative insights prioritized over hypothesis testing.

REFERENCES

Bhandari, K., Roy, A., Wang, K., Puri, G., Colton, S., & Herremans, D. (2025). *Text2MIDI: Generating Symbolic Music from Captions*. In Proceedings of the 39th AAAI Conference on Artificial Intelligence (Vol. 39, No. 22, pp. 23478-23486). DOI:10.1609/aaai.v39i22.34516

Carsault, T., Nika, J., Esling, P., & Assayag, G. (2021). Combining real-time extraction and prediction of musical chord progressions for creative applications. *Electronics, 10*(21), 2634. https://doi.org/10.3390/electronics10212634

de Berardinis, J., Meroño-Peñuela, A., Poltronieri, A., & Presutti, V. (2023). ChoCo: A chord corpus and a data transformation workflow for musical harmony knowledge graphs. *Scientific Data, 10*, 641. https://doi.org/10.1038/s41597-023-02410-w

Le, D.-V.-T., Keller, M., Bigo, L., & Herremans, D. (2025). Natural language processing methods for symbolic music generation and information retrieval: A survey. *ACM Computing Surveys, 57*(7), Article 175. https://doi.org/10.1145/3714457

Liu, C., Wang, H., Zhao, J., Zhao, S., Bu, H., Xu, X., … Qin, Y. (2025). MusicEval: A generative music dataset with expert ratings for automatic text-to-music evaluation. In *Proceedings of IEEE ICASSP 2025* (pp. 4813–4817). https://doi.org/10.1109/ICASSP49660.2025.10890307

Muhamed, A., Li, L., Shi, X., Yaddanapudi, S., Chi, W., Jackson, D., … Smola, A. (2021). Symbolic music generation with Transformer-GANs. In *Proceedings of the 35th AAAI Conference on Artificial Intelligence* (pp. 408–417). https://doi.org/10.1609/AAAI.v35i1.16117

Raposo, A. N., & Soares, V. N. G. J. (2025). Generative jazz chord progressions: A statistical approach to harmonic creativity. *Information, 16*(6), 504. https://doi.org/10.3390/info16060504

Sarmento, P., Kumar, A., Chen, Y.-H., Carr, C. J., Zukowski, Z., & Barthet, M. (2023). GTR-CTRL: Instrument and genre conditioning for guitar-focused music generation with Transformers. In A. Liapis et al. (Eds.), *Artificial Intelligence in Music, Sound, Art and Design – EvoMUSART 2023* (pp. 260–275). Springer. https://doi.org/10.1007/978-3-031-29956-8_17

Wu, S., Yang, Y., Wang, Z., Li, X., & Sun, M. (2024). Generating chord progression from melody with flexible harmonic rhythm and controllable harmonic density. *EURASIP Journal on Audio, Speech, and Music Processing, 2024*(1), Article 4. https://doi.org/10.1186/s13636-023-00314-6

Zhao, Y., Yang, M., Lin, Y., Zhang, X., Shi, F., Wang, Z., … Ning, H. (2025). AI-enabled text-to-music generation: A comprehensive review of methods, frameworks, and future directions. *Electronics, 14*(6), 1197. https://doi.org/10.3390/electronics14061197

RESEARCH SCHEDULE

PHASE	OBJECTIVES	TIMELINE
Proposal Finalization	Submit approved proposal	Oct–Nov 2025
Dataset Preparation	Collect GuitarSet and MIDI/tab data; annotate 150 text–chord–strum triples; quality control	Nov 2025–Jan 2026
Prompt Parser Development	Fine-tune DistilBERT for feature extraction; validate on held-out samples	Dec 2025–Feb 2026
Sequence Model Implementation	Implement Transformer encoder–decoder; train on annotated dataset; conduct ablation studies	Jan–Feb 2026
Rule-Based Fallback & Baseline Testing	Code template-based system; run baseline comparisons with ChatGPT and rule-based outputs	Feb 2026
Objective Evaluation	Compute chord correctness, rhythmic consistency, stylistic diversity; perform error analysis	Early Mar 2026
User Study	Recruit 10 guitarists; conduct remote evaluation; collect ratings and feedback; analyze results	Early Mar 2026
Thesis Writing & Revision	Compile results; write and revise chapters; integrate feedback; finalize references and appendices	Mar 2026
Submission	Final proofreading and submission.	End of Feb–1st week Mar 2026